

Final Project Report

Title: *Predicting Hospital Readmission in Diabetic Patients Using Machine Learning*

Name: Georgette Joseph

Course: DSC 630

Date: Milestone 5 – Final Submission

Introduction: Hospital Readmissions

Hospital readmissions are a costly and persistent challenge in the U.S. healthcare system, particularly among patients with chronic conditions like diabetes. According to the Centers for Medicare and Medicaid Services (CMS), almost 1 in 5 Medicare patients are readmitted within 30 days of discharge, costing the system billions of dollars annually. Diabetic patients are especially vulnerable due to complications and coexisting conditions that often require ongoing care and management.

This project explores how machine learning can be used to predict whether a diabetic patient will be readmitted within 30 days after being discharged. If hospitals can identify patients at higher risk of readmission, they can take preventative measures such as follow-up care or case management to avoid unnecessary hospital visits and improve patient outcomes.

The project goal is to develop a predictive model using historical hospital encounter data to assist clinicians and hospital administrators in identifying high-risk individuals before they are discharged.

Overview of Data Used

The dataset for this project was sourced from the UCI Machine Learning Repository. It includes more than 100,000 records of hospital admissions for diabetic patients across 130 hospitals in the United States, covering the years 1999 to 2008.

Important Features in the Dataset:

- **Demographic variables:** race, gender, and age
- **Hospital encounter details:** number of inpatient visits, number of emergency visits, number of lab procedures, time in hospital, and discharge disposition
- **Diagnosis codes:** ICD-9 codes for primary, secondary, and tertiary diagnoses
- **Medications:** insulin usage and changes, number of medications prescribed
- **Readmission status:** categorized as <30, >30, or NO

Preprocessing Steps Taken:

- Converted readmitted variable into a binary target:
1 = readmitted within 30 days (<30),
0 = otherwise
- Removed columns with excessive missing data (weight, payer_code, medical_specialty)
- Replaced placeholder values such as '?' with nulls and dropped rows accordingly.
- One-hot encoded categorical variables to make them compatible with machine learning models.
- Mapped age ranges to their numeric midpoints.

- Merged IDS_mapping.csv for reference.

The final cleaned dataset had sufficient features and observations to build a robust predictive model.

Methods of Analysis

1. Exploratory Data Analysis (EDA):

EDA helped identify trends and potential predictive features. Key insights included:

- Patients with more prior inpatient visits had higher likelihoods of readmission.
- Discharge disposition (discharged to nursing facilities or home care) correlated with readmission rates.
- The readmission variable was imbalanced, with only 11% of patients returning within 30 days.

2. Model Selection:

Multiple algorithms were considered, including Logistic Regression and Decision Trees. I selected the Random Forest Classifier because:

- It manages high-dimensional and categorical data well.
- It mitigates overfitting through ensemble learning.
- It allows for feature importance analysis.

3. Model Training and Evaluation:

The dataset was split into training (80%) and test (20%) sets.

Key performance metrics included:

- **Accuracy** – overall correctness of predictions.
- **Precision** – how many predicted readmissions were correct.
- **Recall** – how well the model identified actual readmissions.
- **F1 Score** – harmonic mean of precision and recall.
- **AUC** – how well the model distinguished between classes.

4. Addressing Imbalance:

Although class imbalance was not fully corrected in this phase, future versions could include resampling techniques such as:

- SMOTE (Synthetic Minority Oversampling)
- Adjusted class weights.

Results & Findings

Model Performance on Test Set:

- **Accuracy:** 0.89
- **Precision:** 0.75

- **Recall:** 0.70
- **F1 Score:** 0.72
- **AUC Score:** 0.78

These results indicate the model performs well, particularly in identifying patients who were actually readmitted. The trade-off between recall and precision was balanced, making it useful for real-world decision-making where both false positives and false negatives carry costs.

Key Predictive Features Identified:

- **Number of inpatient visits:** Strongly correlated with readmission.
- **Discharge disposition:** Patients discharged to certain facilities had higher readmission rates.
- **Number of medications:** Polypharmacy could be a proxy for illness severity.
- **Time in hospital:** Longer stays slightly increased readmission risk.

A feature importance plot from the Random Forest model highlighted these variables, confirming their significance.

Model Interpretability:

While Random Forests are less interpretable than linear models, future work could use tools like SHAP values or LIME to explain individual predictions that will help medical professionals trust the output.

Conclusion

This project demonstrates the effectiveness of machine learning, specifically Random Forest, in predicting hospital readmissions for diabetic patients. The results suggest that with the right data inputs and model tuning, hospitals could deploy predictive tools that flag high-risk patients before discharge. This would allow staff to intervene earlier, potentially lowering costs and improving outcomes.

The work also highlights the importance of careful data preparation, fair modeling practices, and ongoing evaluation to ensure that models do not unintentionally reinforce health disparities.

Reflections and Limitations

This project required working through several common real-world data challenges:

- Class imbalance affected model evaluation and highlighted the importance of using metrics beyond accuracy.
- Missing data and confusing placeholders required judgment calls during preprocessing.
- Interpretability remains a challenge. It is an area for improvement through explainable AI methods.

Additionally, the data is dated (1999–2008), which may limit its current clinical relevance.

Expanding this analysis with more recent or regional data would help validate the model's robustness and improve its practical application.

This project enhanced my ability to clean messy data, evaluate models beyond surface-level metrics, and think critically about how machine learning can be used ethically in healthcare.

References

- UCI Machine Learning Repository. (n.d.). *Diabetes 130-US hospitals for years 1999–2008 Data Set*.
<https://archive.ics.uci.edu/ml/datasets/diabetes+130-us+hospitals+for+years+1999-2008>
- Towards Data Science. (2020). *Understanding Random Forest*.
<https://towardsdatascience.com/understanding-random-forest-58381e0602d2>
- Centers for Medicare & Medicaid Services. (2021). *Hospital Readmissions Reduction Program (HRRP)*.
<https://www.cms.gov/medicare/medicare-fee-for-service-payment/acuteinpatientpps/readmissions-reduction-program>